

A Weighted Compatibility Algorithm for Community Education Volunteer Coordination: Design, Simulation, and Validation

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Abstract—Coordinating educational volunteers in vulnerable urban communities is a persistent challenge due to the manual, unstructured nature of current assignment processes, which fail to account for skill compatibility, schedule availability, and geographic constraints simultaneously. This paper presents the design and computational validation of a Community Education Volunteer Coordination Platform, whose core is a weighted compatibility algorithm $C(v, p) = 0.40 \cdot \text{Skill} + 0.30 \cdot \text{Schedule} + 0.15 \cdot \text{Location} + 0.15 \cdot \text{Experience}$, derived from empirical survey data collected from 50 potential volunteers at Universidad Distrital Francisco José de Caldas. Simulation results from a discrete-event model and a cellular automata engagement model confirm that the algorithm achieves mean compatibility scores of 0.860–0.887 across all tested scenarios, project coverage reaches 100% when the volunteer pool exceeds 30 participants, and the platform produces an 18% relative increase in community engagement while revealing volunteer burnout contagion as a previously unidentified operational risk.

Index Terms—volunteer coordination, matching algorithm, discrete-event simulation, cellular automata, community education, systems design

I. INTRODUCTION

Access to academic support in socioeconomically vulnerable communities remains an open problem in many Latin American cities. In Bogotá D.C., numerous schools serving low-income populations rely on external volunteers to provide tutoring and mentoring, yet the process of connecting willing volunteers with institutions that need them is largely manual, inefficient, and difficult to scale [7].

Prior work on volunteer management systems has addressed isolated dimensions of this coordination challenge. Scheduling and resource allocation algorithms inspired by operational research have been applied to humanitarian logistics [4], while

agent-based and cellular automata models have been used to study social contagion and collective action dynamics in community networks [5], [6]. Quality management frameworks such as ISO 9001:2015 [2] and risk management standards such as ISO 31000:2018 [1] provide the methodological scaffolding for engineering robust sociotechnical systems. Project management bodies of knowledge, in particular the PMBOK 7th edition [3], offer structured guidance for lifecycle planning. Despite these contributions, no integrated digital platform specifically designed for educational volunteer coordination in the Latin American context—combining multi-dimensional skill matching, conflict-free scheduling, and empirically grounded simulation validation—has been reported in the literature.

This paper describes the full engineering lifecycle of such a platform, from problem analysis and primary data collection through system design, risk management, and computational simulation. The central contribution is a weighted compatibility algorithm whose parameters are directly calibrated from survey data, and whose performance is validated through two complementary simulation paradigms: a discrete-event model of the assignment pipeline and a cellular automata model of volunteer engagement dynamics. Together, these artifacts provide both process-level and behavior-level evidence that the proposed system meets its design targets and is viable for deployment in real community education settings.

II. METHODS AND MATERIALS

A. Empirical Foundation and Requirements

The design process began with a digital survey administered to 50 potential volunteers at Universidad Distrital Francisco

José de Caldas. Responses revealed that 40% of participants were Engineering students, 22% were from Teaching and Education programs, and 54% could dedicate fewer than two hours per week to tutoring activities. In-person and virtual delivery modalities were preferred equally (36% each), with 28% willing to use both. These findings directly shaped the algorithm weight structure and the system’s emphasis on short, flexible sessions.

Eight functional requirements were elicited from stakeholder analysis, covering volunteer and institution registration, the matching algorithm, conflict-free scheduling, session tracking, report generation, a notification service, and an administration panel. Five non-functional requirements established measurable targets: matching response time below 60 seconds, system availability above 99%, volunteer registration completable in under 5 minutes, support for 80 or more concurrent users, and modular deployability aligned with microservices architecture.

B. System Architecture

The platform follows a three-tier architecture aligned with ISO 9001:2015 modularity principles [2]. The *presentation layer* hosts a Volunteer Portal, an Institution Portal, an Administration Dashboard, and a Reports and Analytics module. The *application layer* contains the Volunteer and Institution Registration modules, the Matching Algorithm Engine, the Scheduling Engine, the Academic Impact Monitor, and a Notification Service. The *data layer* persists state in four dedicated repositories: a Volunteer Profile database, a Projects and Schedules database, a Sessions log, and an Analytics warehouse. For the prototype phase, Google Sheets API serves as a temporary database accessible from Python, with a planned migration to PostgreSQL once the platform exceeds 50 active users.

C. Compatibility Algorithm

The system’s core is the weighted compatibility function:

$$C(v, p) = 0.40 \cdot Sk(v, p) + 0.30 \cdot Sc(v, p) + 0.15 \cdot Sl(v, p) + 0.15 \cdot Se(v, p) \quad (1)$$

where Sk measures skill overlap between the volunteer’s academic competencies and the project’s subject requirements; Sc quantifies the percentage of required time slots covered by the volunteer’s availability windows; Sl encodes geographic proximity for in-person modalities or connectivity quality for virtual ones; and Se maps the volunteer’s prior tutoring experience to the target school level. Weights were set according to survey-validated priorities, with skill compatibility ranked highest by respondents. A score above 0.80 was established as the minimum acceptance threshold for any assignment.

D. Risk Management

Risk identification followed ISO 31000:2018 guidelines [1], producing two registers. Technical risks include poor data quality affecting algorithm accuracy (RT01, probability×impact score: 9, critical), Google Sheets API outages

(RT02), low connectivity in communities (RT03), miscalibrated algorithm weights (RT04), and scheduling conflicts (RT05). Operational risks include volunteer supply fluctuation (RO01, score: 9, critical), low institutional adoption (RO02), high cancellation rates (RO03, estimated at 15% from pilot data), and calendar disruptions (RO04). Each risk was assigned a specific mitigation mechanism, such as a volunteer waiting queue for RO01 and local data caching for RT02.

E. Simulation Models

Two simulation paradigms were implemented in Python with a fixed random seed for reproducibility (SEED = 42).

Discrete-Event Simulation (DES). The assignment pipeline was modeled as a sequence of discrete events—profile submission, matching computation, scheduling, and session confirmation—following standard DES methodology [4]. Three scenarios were tested over a 12-week horizon: Baseline (50 volunteers, 20 projects), Optimistic (80 volunteers, 20 projects), and Stress (25 volunteers, 20 projects). A stochastic volunteer churn of ± 3 per week was introduced to replicate realistic pool fluctuation. Parameters were calibrated directly from the survey data and Workshop 3 risk register (Table I).

Cellular Automata (CA) Engagement Model. A 30×30 grid represented the university community, with each cell cycling through five states: Inactive $\rightarrow$ Aware $\rightarrow$ Engaged $\rightarrow$ Active Tutor $\rightarrow$ Burned Out $\rightarrow$ Inactive. Transition probabilities were governed by Moore neighborhood influence and a platform activation boost. The burnout contagion rule, $P(\text{Active} \rightarrow \text{Burned Out}) = 0.08 + 0.05 \cdot n_{\text{burnout}}$, captures social cascading effects. Two conditions were compared: Platform ON versus Platform OFF.

TABLE I
KEY SIMULATION PARAMETERS

Parameter	Value	Source
Low availability ratio	54%	Survey Q5
Engineering volunteers	40%	Survey Q2
Session cancellation rate	15%	Risk RO03
Algorithm latency (mean)	45 ms	W2 NFR
Skill weight w_{sk}	0.40	W2 design
Schedule weight w_{sc}	0.30	W2 design
Simulation horizon	12 weeks	W3 Gantt
CA grid size	30×30	UD campus
Initial engagement ratio	8%	Survey base

III. RESULTS AND DISCUSSION

A. Discrete-Event Simulation

Table II summarizes the DES outcomes across the three scenarios. All scenarios achieved 100% project coverage and mean compatibility scores well above the 0.80 threshold. Assignment latency remained near 45 ms in all cases, comfortably within the 60-second non-functional requirement.

A counter-intuitive finding emerged from the Stress scenario: the mean compatibility score was highest (0.887) when the volunteer pool was smallest. This quality paradox arises

TABLE II
DES RESULTS SUMMARY

Scenario	Coverage	Mean $C(v, p)$	Cancelled	Target met
Baseline (50 vols.)	100%	0.860	3	Yes
Optimistic (80 vols.)	100%	0.870	4	Yes
Stress (25 vols.)	100%	0.887	1	Yes
<i>Design target (W2)</i> $>70\%$ coverage, $C(v, p) > 0.80$				

from the algorithm’s optimization logic—with fewer candidates, each assignment concentrates on the most compatible pairing rather than distributing to marginally qualified volunteers. Weekly temporal dynamics confirmed that coverage drops below 70% only when the active pool falls under 30 volunteers, empirically confirming the criticality of risk RO01 and validating the 30-volunteer floor as a policy threshold.

Sensitivity analysis further validated three design decisions: (1) latency scales sub-linearly with pool size, remaining below 60 s for all tested configurations up to 100 volunteers; (2) coverage degrades rapidly below the 30-volunteer threshold; and (3) the skill weight $w_{sk} = 0.40$ lies within an optimal plateau of the sensitivity curve, with degradation occurring for values above 0.55 as schedule and location constraints become under-weighted.

B. Cellular Automata Simulation

The CA model revealed two principal emergent behaviors. First, the platform produced a positive network amplification effect: total engagement reached 24.4% with the platform active, compared to 20.7% without it, a relative increase of 18%. This improvement is not programmed into any single agent—it emerges from compounding local probability boosts propagating through the social network grid. Critical mass formation was observable by week 8, when platform conditions showed denser clusters of active tutors, consistent with a non-linear tipping point near 15% neighborhood activation.

Second, the CA model uncovered a burnout contagion risk that was absent from all prior risk registers. Spatially clustered burnout, plausible when high-demand projects concentrate in the same geographic zone, creates neighborhood effects that accelerate further withdrawal. This cascade pattern follows the Turing morphogenesis dynamics described in complexity theory literature [5]. Despite sensitive dependence on initial conditions in the first 5–8 simulation weeks, all runs converged to a stable attractor region of approximately 140–180 active tutors and 55–75 burned-out agents, confirming macro-level predictability compatible with a complex adaptive systems framework [6].

Table III consolidates the full design validation, mapping each Workshop 2 and Workshop 3 target to its simulation outcome.

All seven pre-specified design targets were validated. The newly identified burnout contagion risk was incorporated into the risk register as RO05 and addressed through a recommended Burnout Monitoring Module that tracks session load per volunteer and triggers alerts above three sessions per week.

TABLE III
DESIGN VALIDATION AGAINST WORKSHOP TARGETS

Design decision	Target	Simulation result	Status
Response time	<60 s	≈ 45 ms	Validated
Project coverage	>70%	100% (all)	Validated
Mean $C(v, p)$	>0.80	0.860–0.887	Validated
Cancellation resilience	Absorb 15%	Unaffected	Validated
Volunteer floor	≥ 30	Drop at <30	Confirmed
Weight $w_{sk} = 0.40$	Optimal quality	Plateau 0.35–0.50	Validated
Platform social effect	Positive recruit.	+18% engagement	Validated
Burnout contagion	Not modeled	Cascade detected	New risk

IV. CONCLUSIONS

This paper presented the full engineering lifecycle of a Community Education Volunteer Coordination Platform, from empirical data collection through algorithm design, risk management, and dual-paradigm computational simulation. The compatibility algorithm $C(v, p)$ met all performance and quality targets across baseline, optimistic, and stress operational scenarios, confirming the validity of the survey-derived weight structure and the three-tier modular architecture. The 30-volunteer minimum threshold emerged as a critical operational policy backed by quantitative evidence, and the platform’s social activation function was independently confirmed to produce an 18% engagement gain through emergent network dynamics.

The cellular automata simulation contributed the most novel insight of the project: volunteer burnout propagates as a social contagion whose cascade risk is invisible to component-level analysis but becomes apparent when the system is studied as a complex adaptive system. This finding demonstrates the value of behavior-oriented simulation alongside traditional process-oriented modeling in sociotechnical system design. Future work should implement the proposed Burnout Monitoring Module, conduct a real-world pilot with partner institutions in Bogotá, recalibrate algorithm weights using post-deployment satisfaction data, and explore geographic load-balancing constraints to prevent spatially clustered burnout cascades.

ACKNOWLEDGMENTS

The authors thank Professor Carlos Andrés Sierra, M.Sc., of the Systems Analysis & Design course at Universidad Distrital Francisco José de Caldas, for his methodological guidance throughout the four workshops that grounded this research.

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